

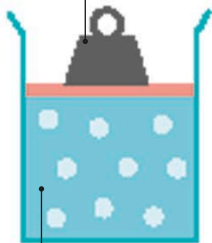
1662

Boyle's law

Anglo-Irish scientist Robert Boyle published a law to show that the volume of a gas decreases with increasing pressure, and vice versa. Boyle carried out many experiments using an air pump he had made. Often called the father of modern chemistry, he introduced the idea of an element as a substance that cannot be broken down.

Diffusion

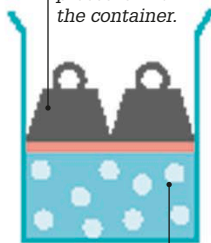
One weight creates pressure within the container.



Evenly spread molecules

Pressure

Two weights double the pressure within the container.



Higher pressure packs the molecules into half of the original space.

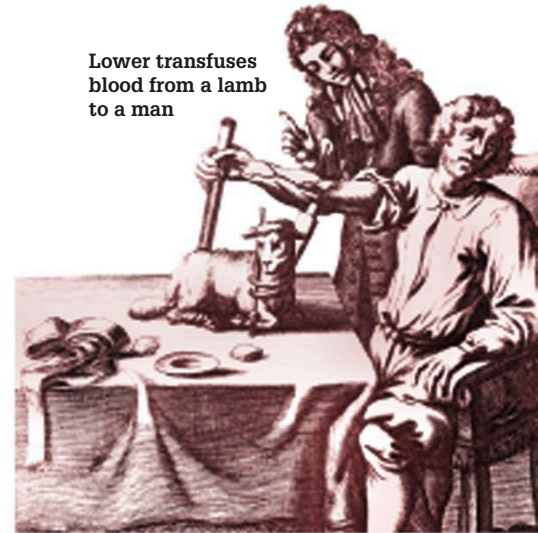


Artist's impression of a dodo

Dead as a dodo

The dodo, a large flightless bird, had become extinct by 1662. It lived only on the island of Mauritius in the Indian Ocean, where it was hunted for food by visiting European sailors.

Lower transfuses blood from a lamb to a man



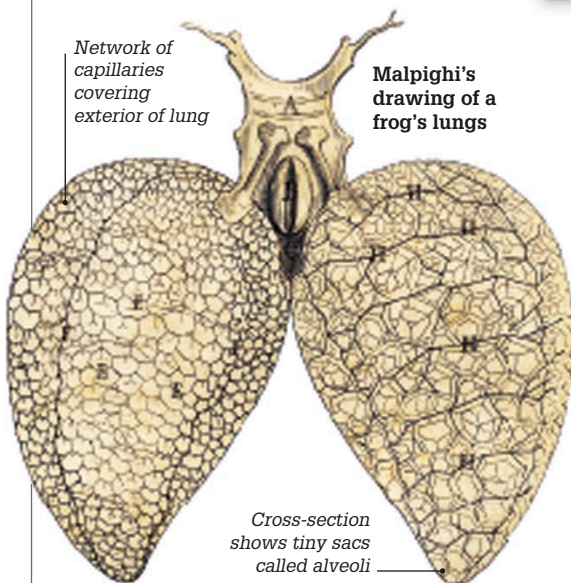
1666

Blood transfusion

Richard Lower, an English doctor, carried out a successful blood transfusion between two dogs. A year later, he injected a small quantity of lamb's blood into the vein of a human patient, who appears to have survived despite the risk of a massive allergic reaction.

1665

1670



Network of capillaries covering exterior of lung

Malpighi's drawing of a frog's lungs

Cross-section shows tiny sacs called alveoli

1661

Microscopic discoveries

Italian biologist Marcello Malpighi used a microscope to study the structure of the lung. He discovered capillaries, the minute blood vessels that connect the veins and arteries. William Harvey had suggested 30 years earlier that these blood vessels existed (see p.75).

1669

Fossils in rock

Danish geologist Nicolas Steno explained that sediment formed horizontal layers of rock (called strata) over time. As new strata formed on top, animal remains within each layer turned gradually into fossils. So the oldest fossils are always at the bottom, with the newer ones above them.



Hennig Brand is said to have used 1,250 gallons (5,670 liters) of human urine in his experiments.

1669

A new element

In his search for the "philosopher's stone," to turn base metal into gold, German alchemist Hennig Brand boiled up concentrated human urine and accidentally discovered the element phosphorus, which gives off a greenish glow.



Hennig Brand in his laboratory